Bioinformatic approaches to regulatory genomics and epigenomics

376-1347-00L | week 06

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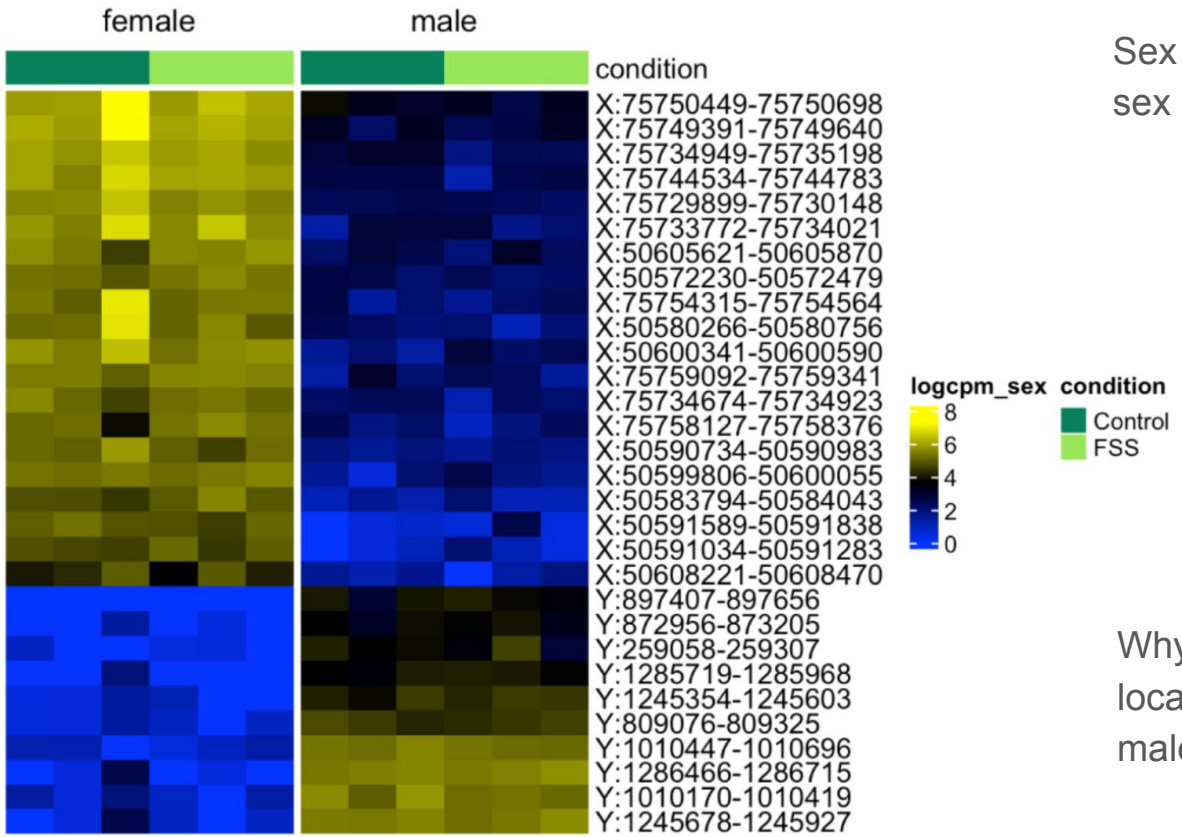
ETH Zürich

Plan

- Debriefing on last week’s assignment
- Overview of transcription factors and their binding specificity
- DNA motifs and related analysis

Debriefing on the assignment



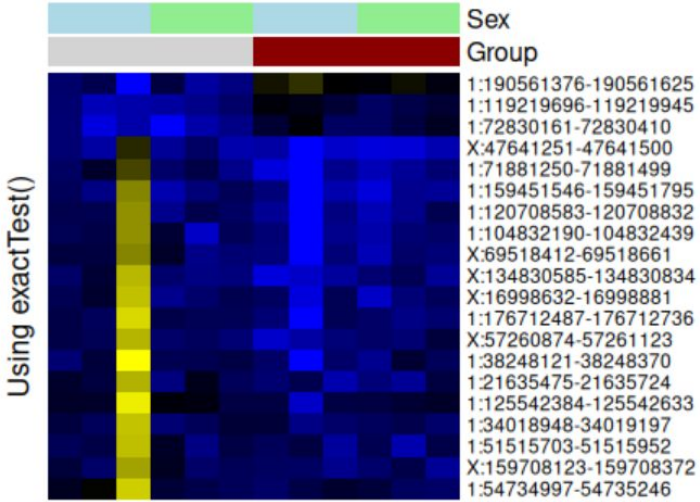
Sex differences are chiefly on the sex chromosomes

Why are the values in regions located on chrY not all zeros for males?

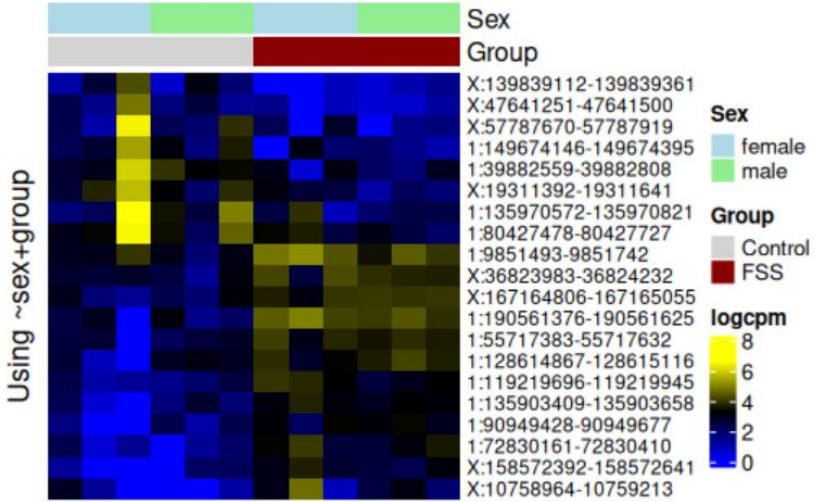
Debriefing on the assignment

Even when you're not interested in Sex, if Sex has an impact, you have to include it in the model! i.e. `~sex+group`

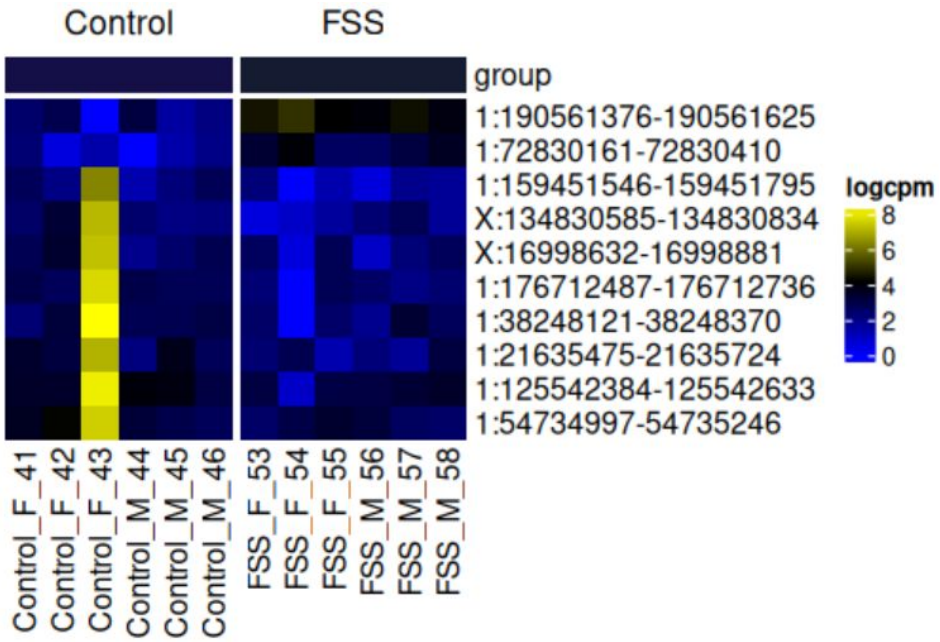
When not including Sex:



Including Sex in the model:



Debriefing on the assignment



Depending on how you made your (stress) test, there seems to be an **outlier**.

However we **cannot** assess this based on the regions coming out of the differential analysis

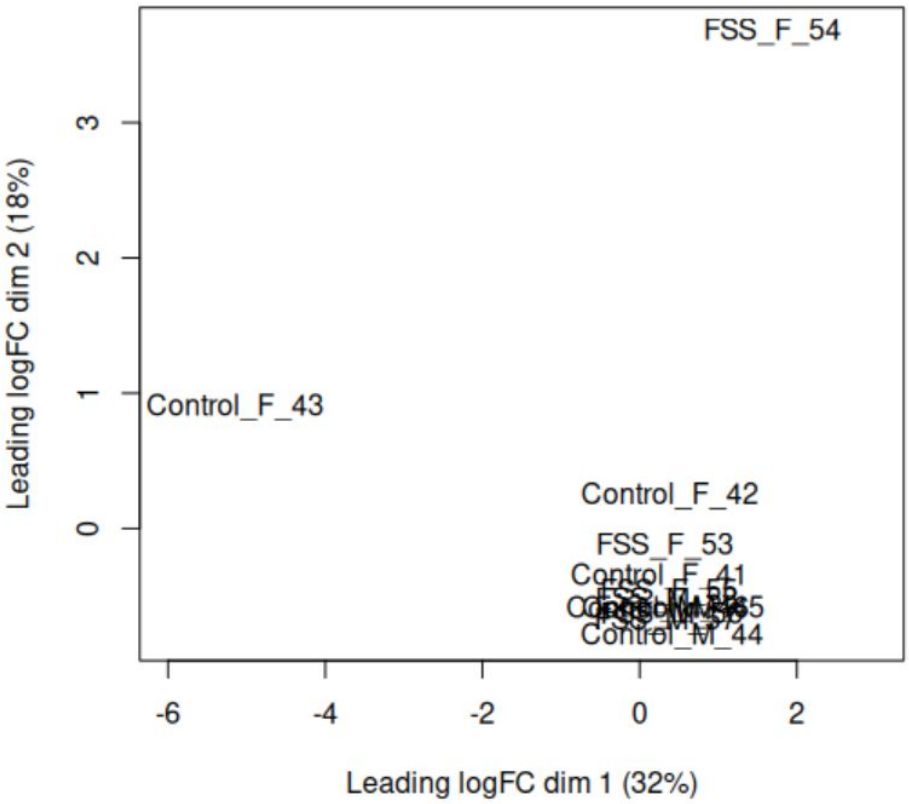
Our options are:

1. First check **quality controls** (e.g. fastqc) in case there's something technical clearly wrong about this sample (there isn't)
2. This is ATAC-seq data, so we could try methods to correct its specific biases -- we'll discuss this after the Easter break.
3. We could look at an overview of the data that's not dependent on experimental groups, e.g. **PCA / MDS**
4. Use surrogate variable analysis (**SVA**) to correct the bias

Debriefing on the assignment

A design-agnostic, multi-dimensional scaling (MDS) plot shows two clear outliers

```
dds <- DGEList(assay(se))
dds <- calcNormFactors(dds)
plotMDS(dds)
```

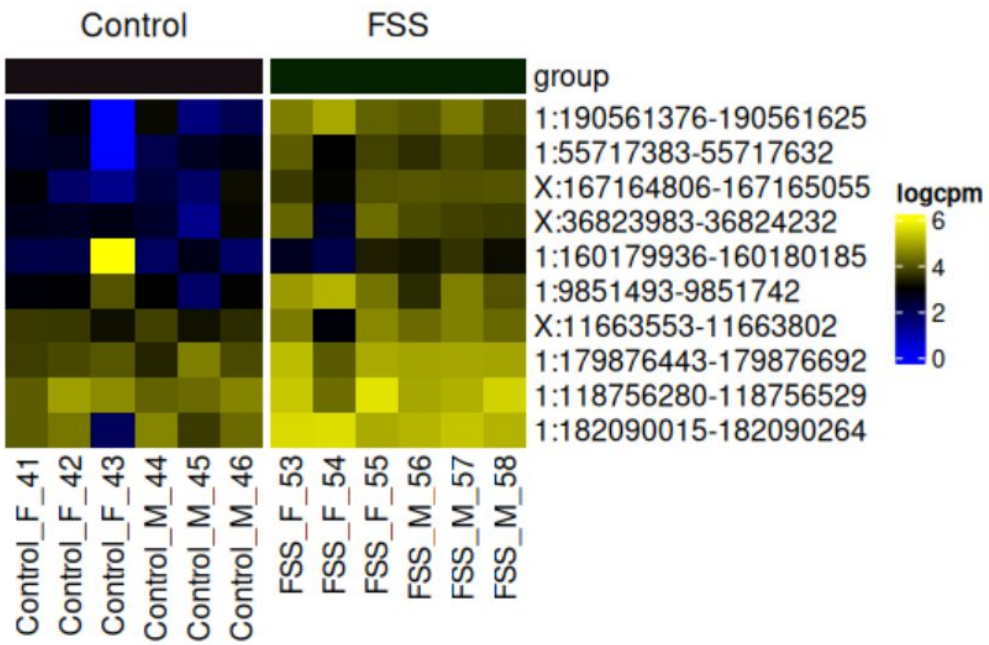


Using SVA

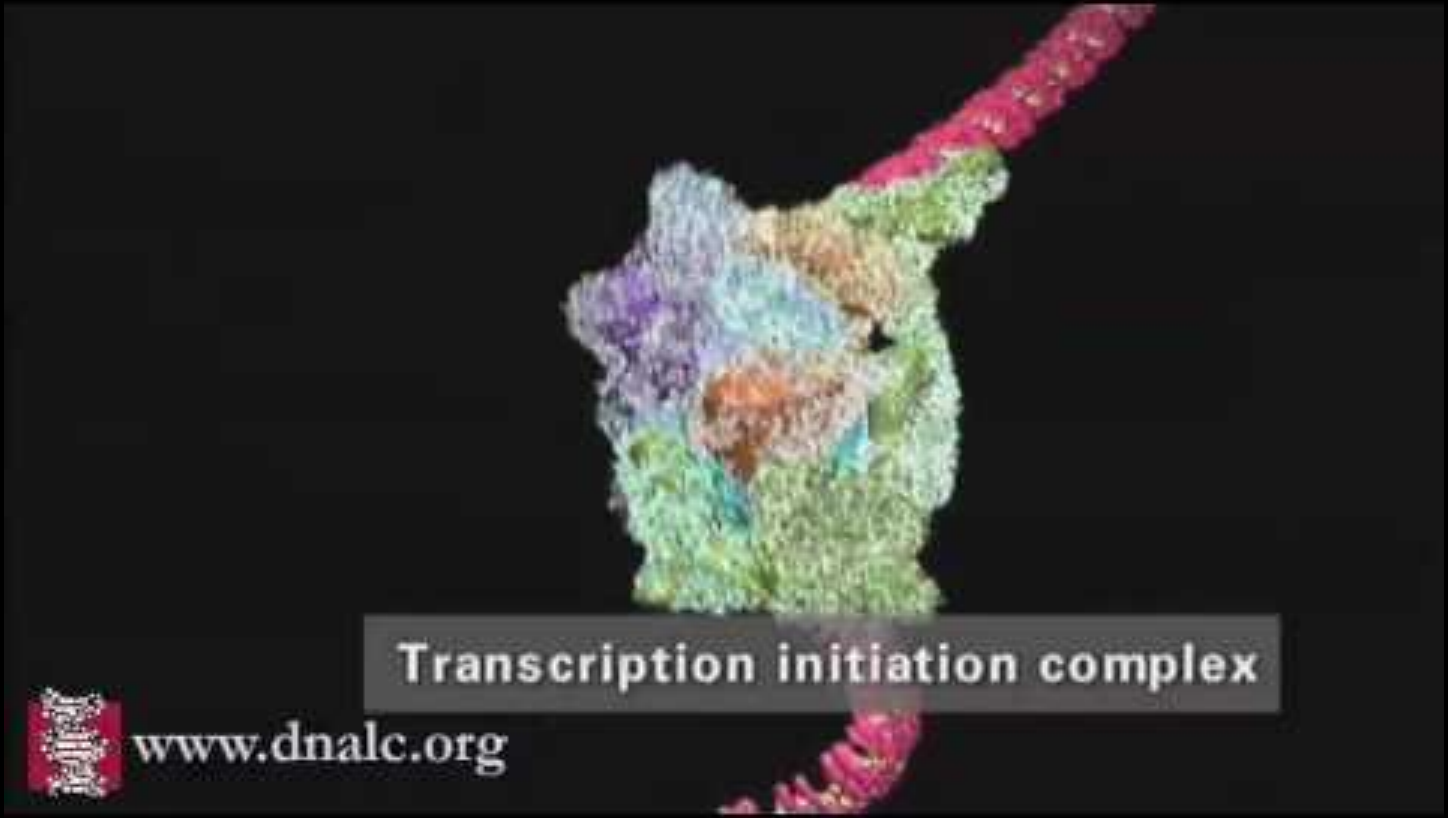
Surrogate-variable analysis (SVA) and derivatives thereof can be used to identify technical vectors of variation, and correct for them in the differential analysis model

```
> se <- SEtools::svacor(se, ~group)
Number of significant surrogate variables is:  2

> mm <- model.matrix(~SV1+SV2+group,
data=as.data.frame(colData(se)))
> dds <- estimateDisp(dds, mm)
```



Transcription factors

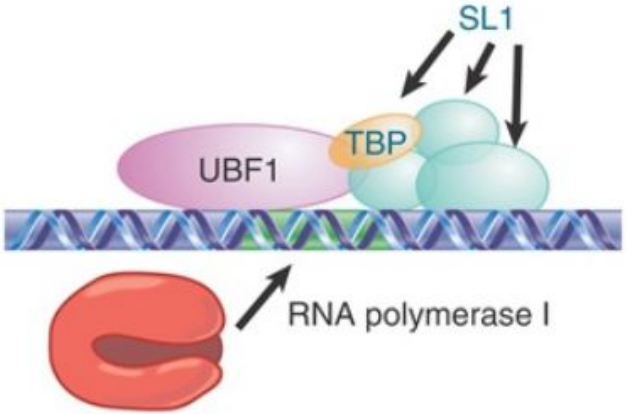


https://youtu.be/SMtWvDbfHLo

(See also https://youtu.be/WW9lIYM_FC0)

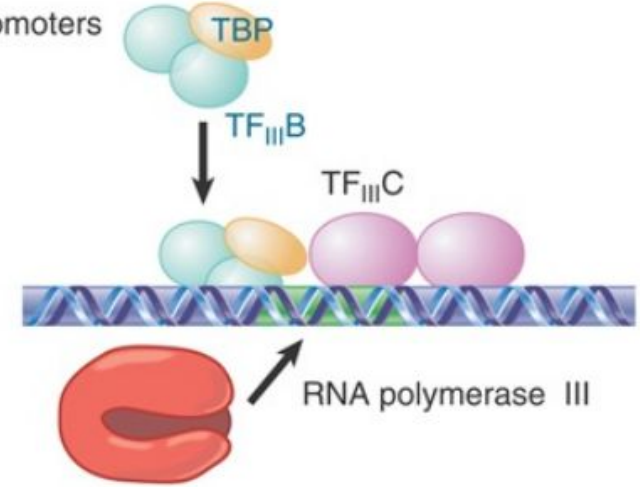
Pol I promoters

rRNA



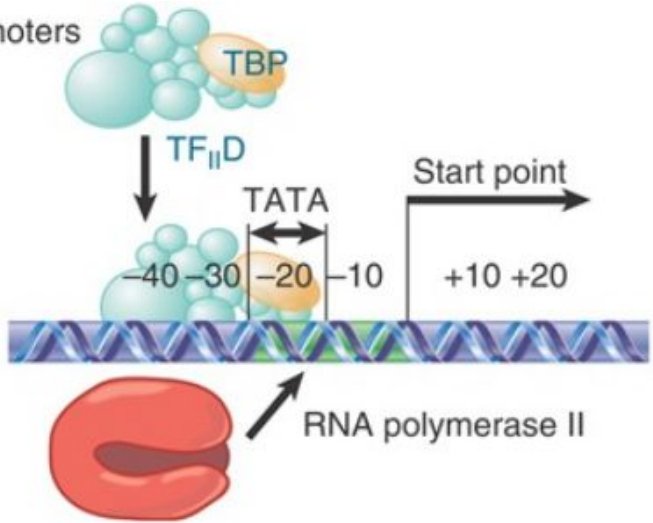
Pol III promoters

tRNA



Pol II promoters

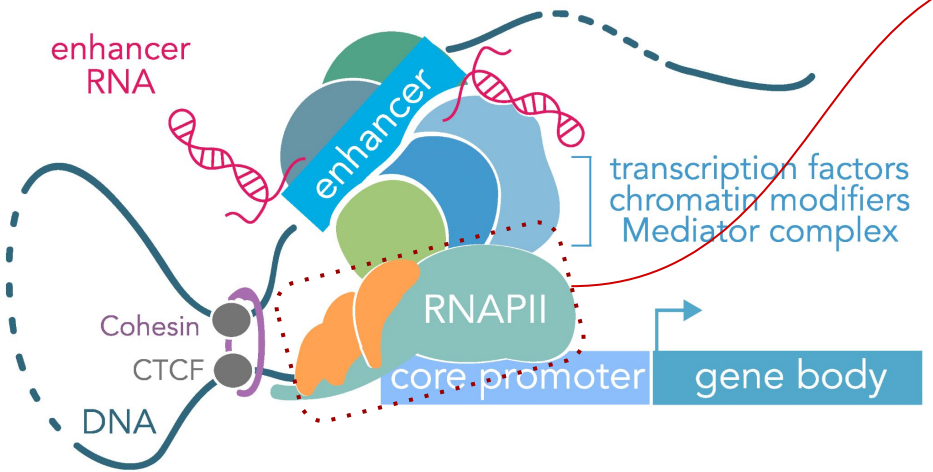
Most RNAs



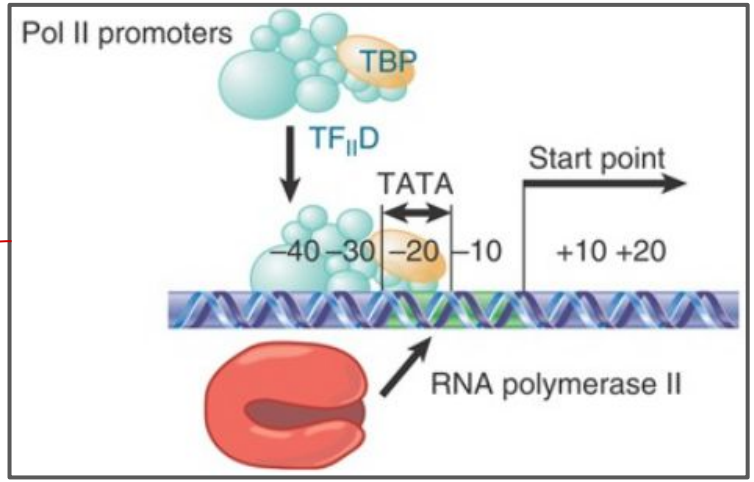
(Adapted from Krebs, Goldstein and Kilpatrick, Genes XII, 2018)

Additional regulatory elements

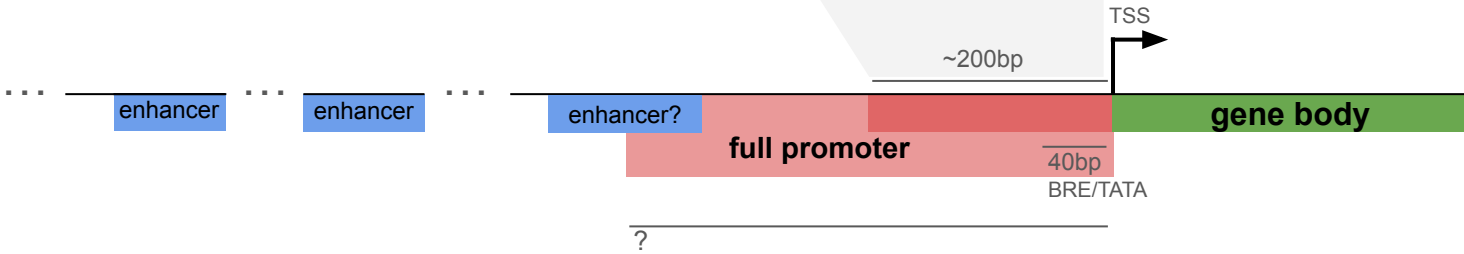
Enhancer-driven gene regulation



(Carullo and Day, Genes 2019)



"function as non-cell-type-specific 'on switches' providing similar expression levels to their associated gene" (Agarwal et al., biorxiv 2023)



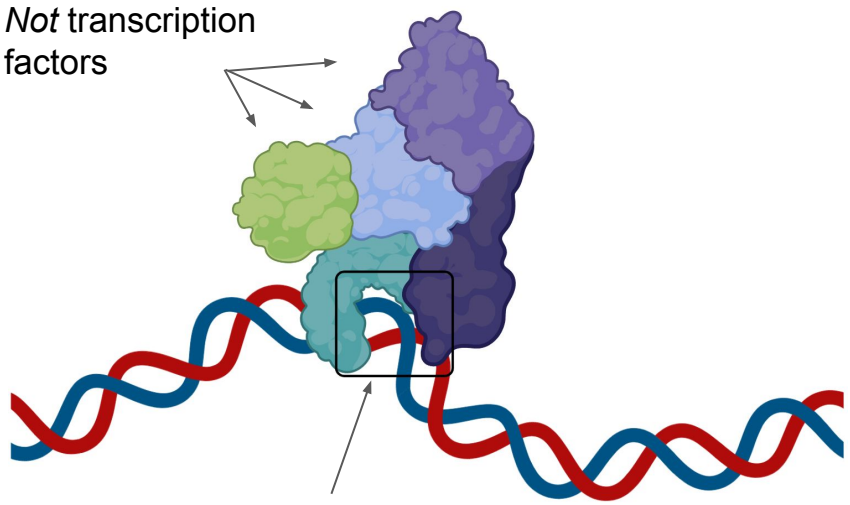
What is a transcription factor?

Proteins capable of both:

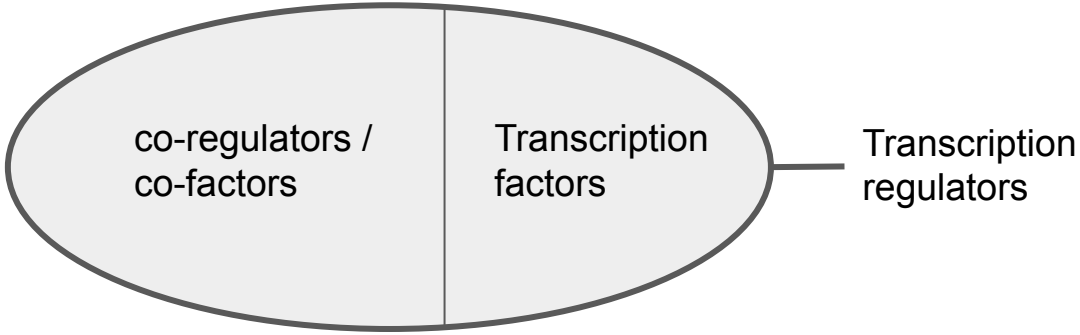
1) Binding DNA in a sequence-specific manner

2) Regulating transcription

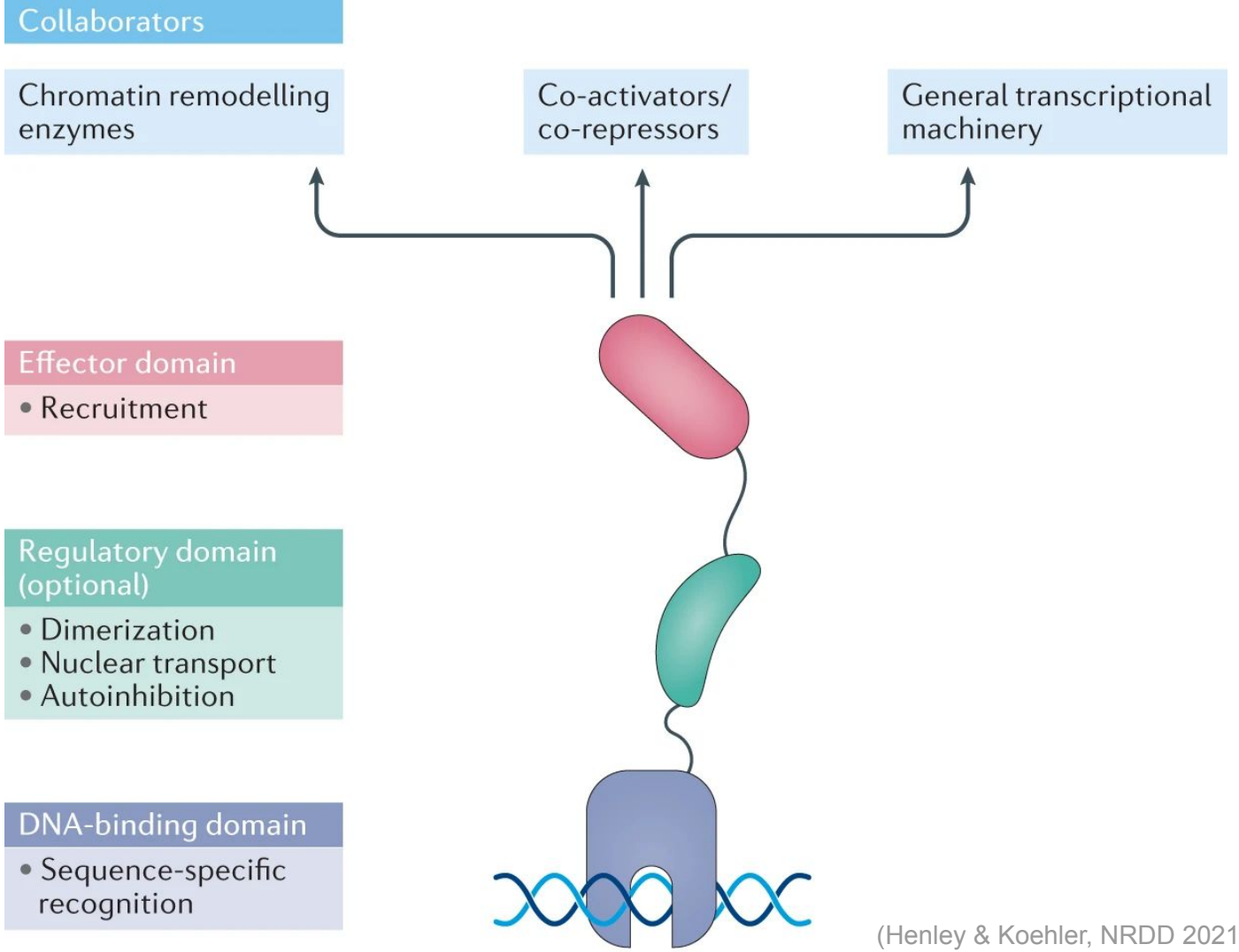
(Lambert et al., Cell 2018)



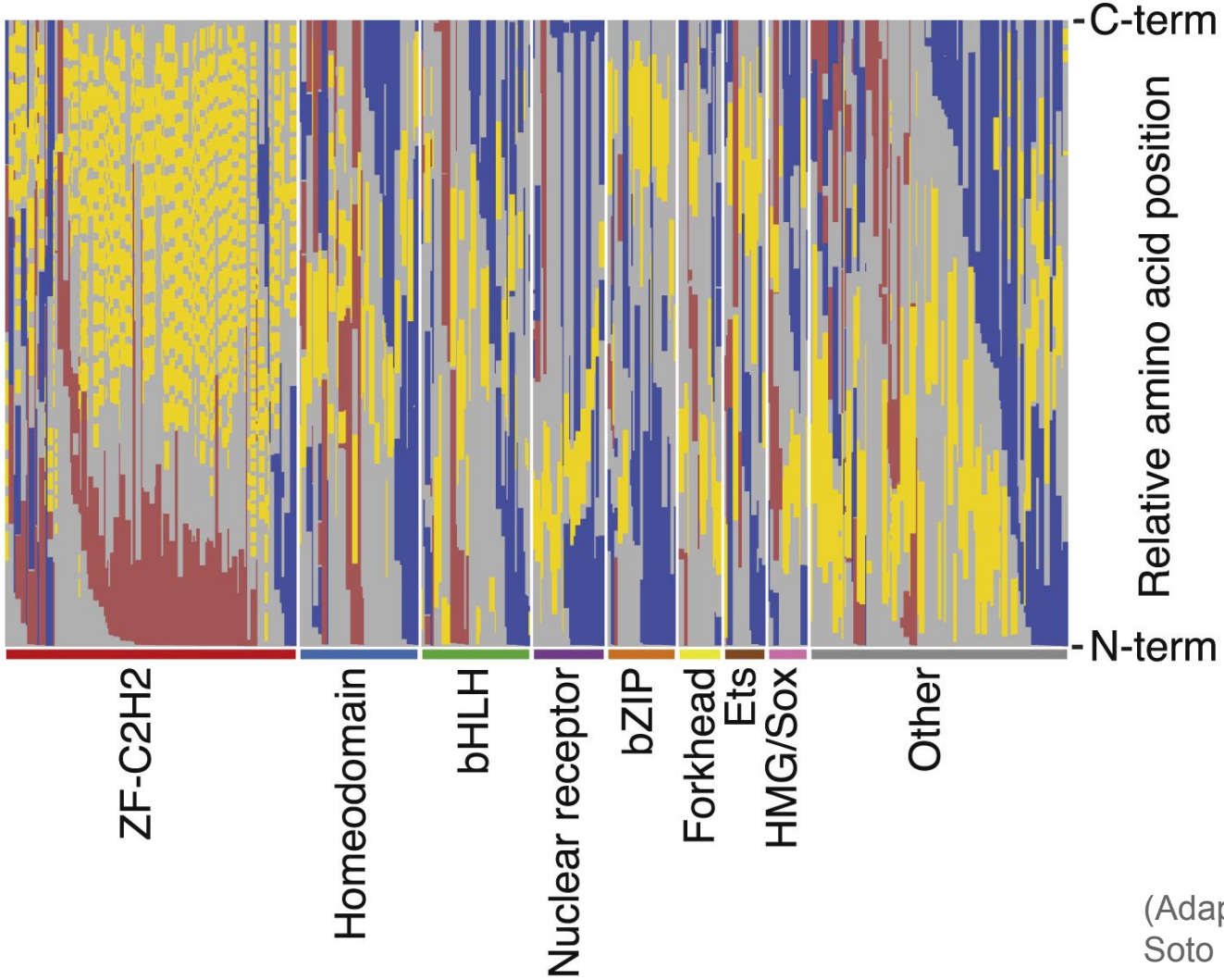
(biorender)



Anatomy of a transcription factor (TF)



(Henley & Koehler, NRDD 2021)



While most TF have either an activating (AD) or repressive (RD) domain, some have both

Domain
- AD
- RD
- DBD
- Other

(Adapted from Soto et al., Molecular Cell 2021)

Cell

(Cell 2018)

Leading Edge
Review

The Human Transcription Factors

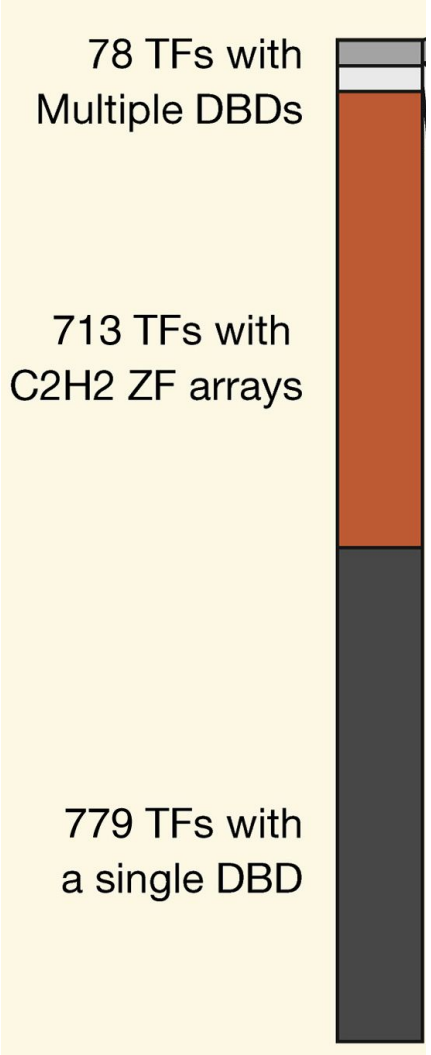
Samuel A. Lambert,[1,9] Arttu Jolma,[2,9] Laura F. Campitelli,[1,9] Pratyush K. Das,[3] Yimeng Yin,[4] Mihai Albu,[2] Xiaoting Chen,[5] Jussi Taipale,[3,4,6,*] Timothy R. Hughes,[1,2,*] and Matthew T. Weirauch[5,7,8,*]

Proteins capable of both:

1) Binding DNA in a sequence-specific manner

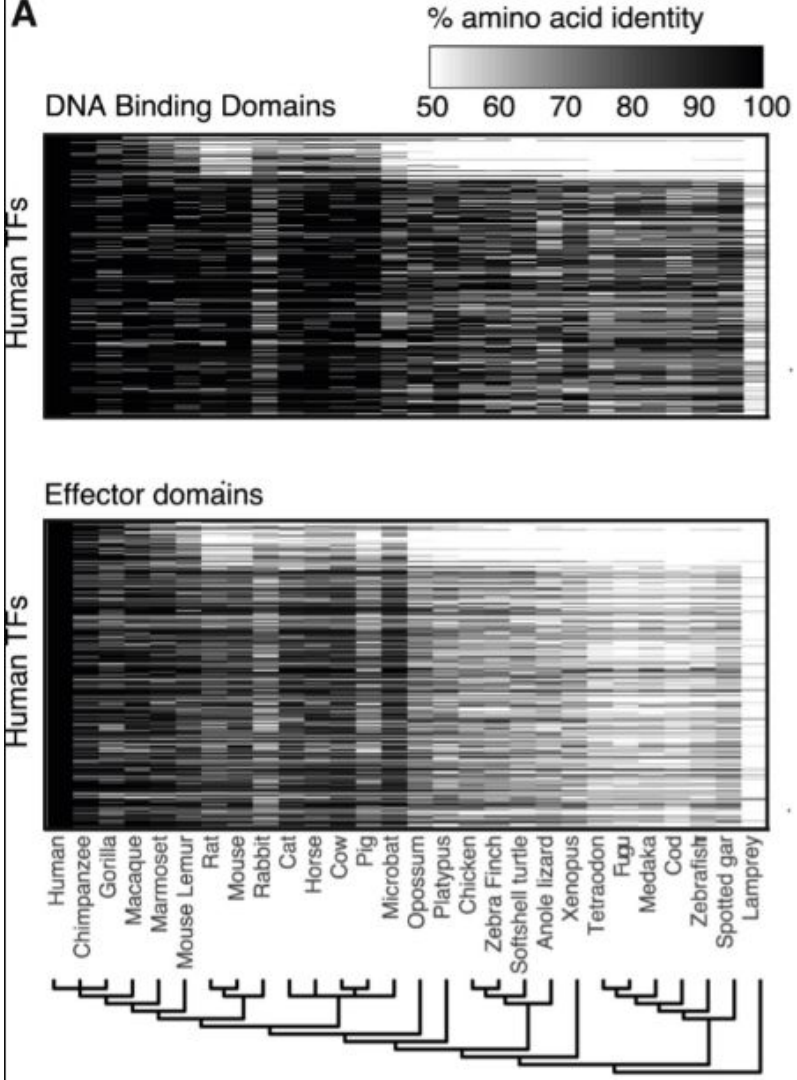
2) Regulating transcription

According to their census, humans have 1570 transcription factors

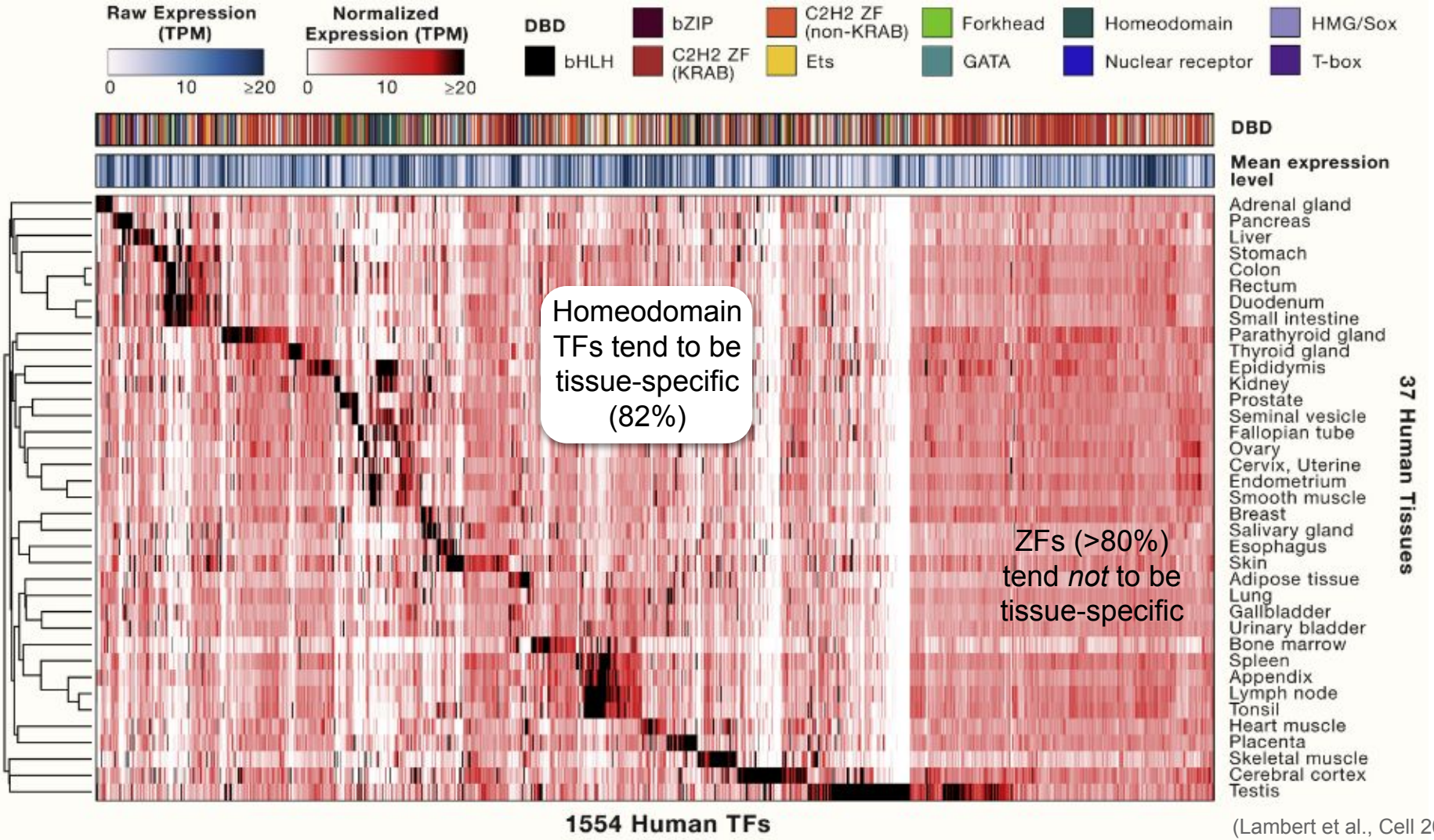


Transcription factors are highly conserved

DNA binding domains show much higher conservation than effector domains



(Soto et al., Molecular Cell 2021)

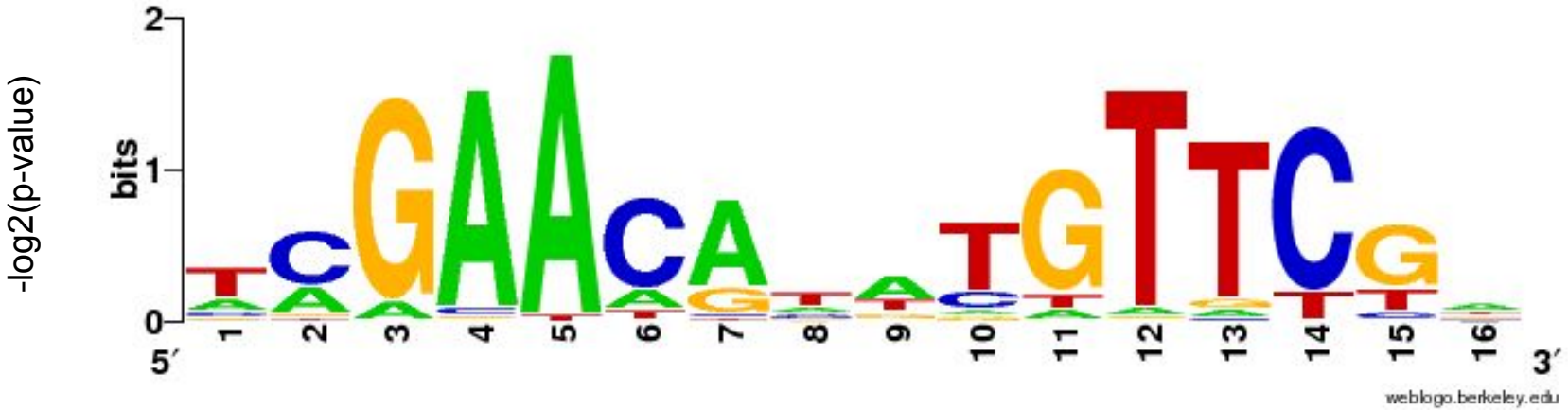


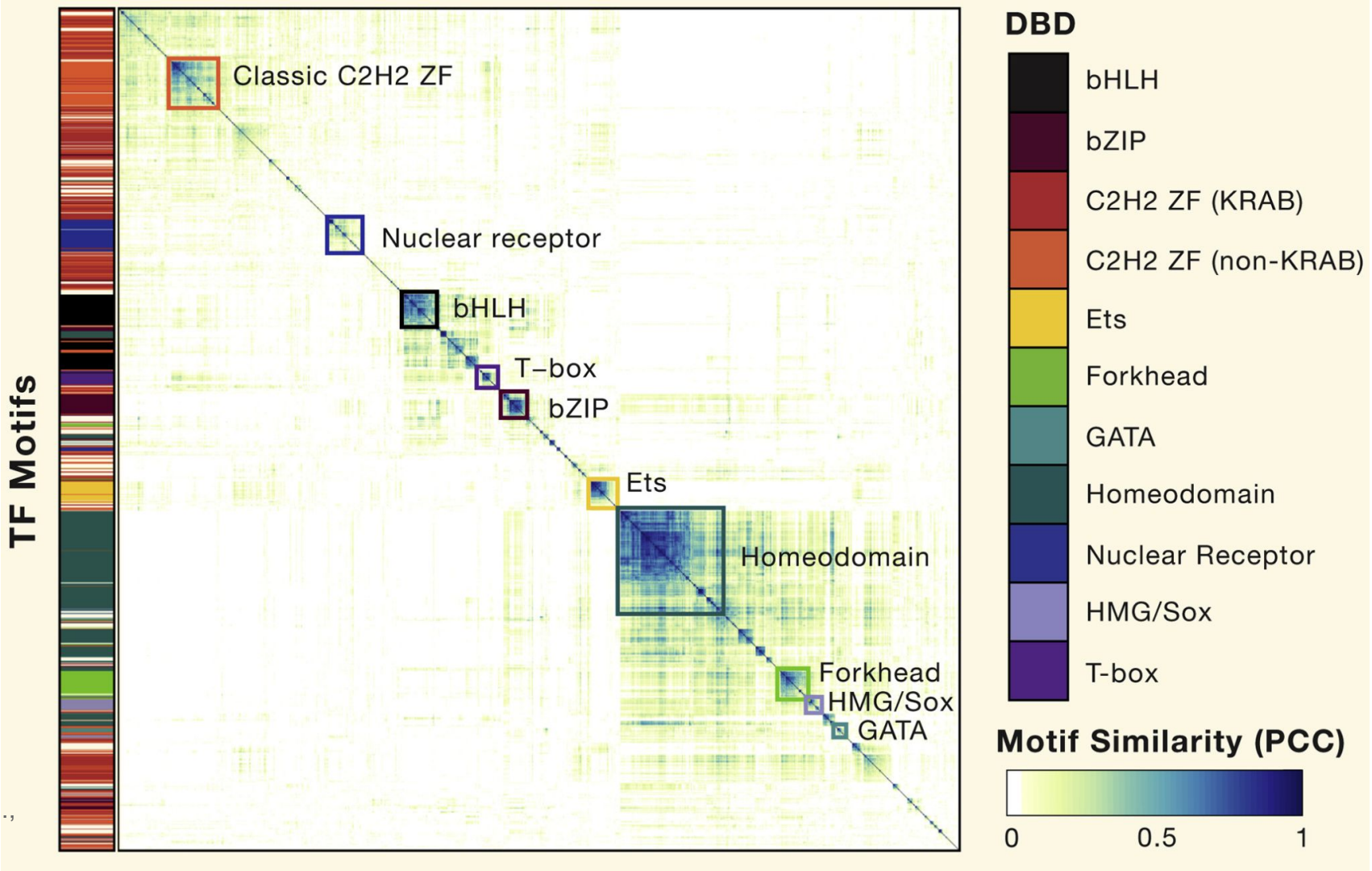
(Lambert et al., Cell 2018)

Sequence-specificity

E.g. The LexA bacterial TF recognizes the consensus sequence

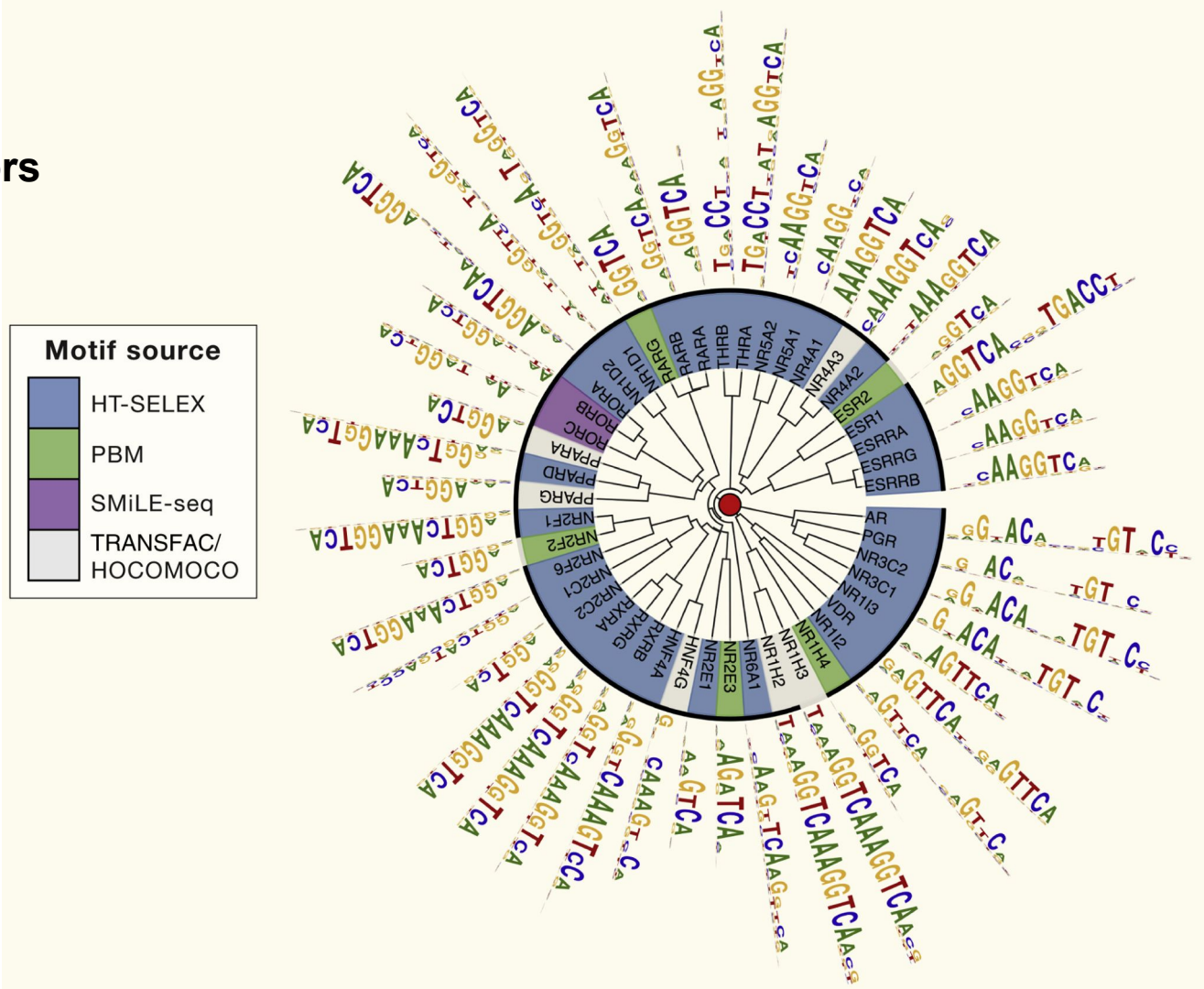
5'-GAACAnnTGTTC-3'





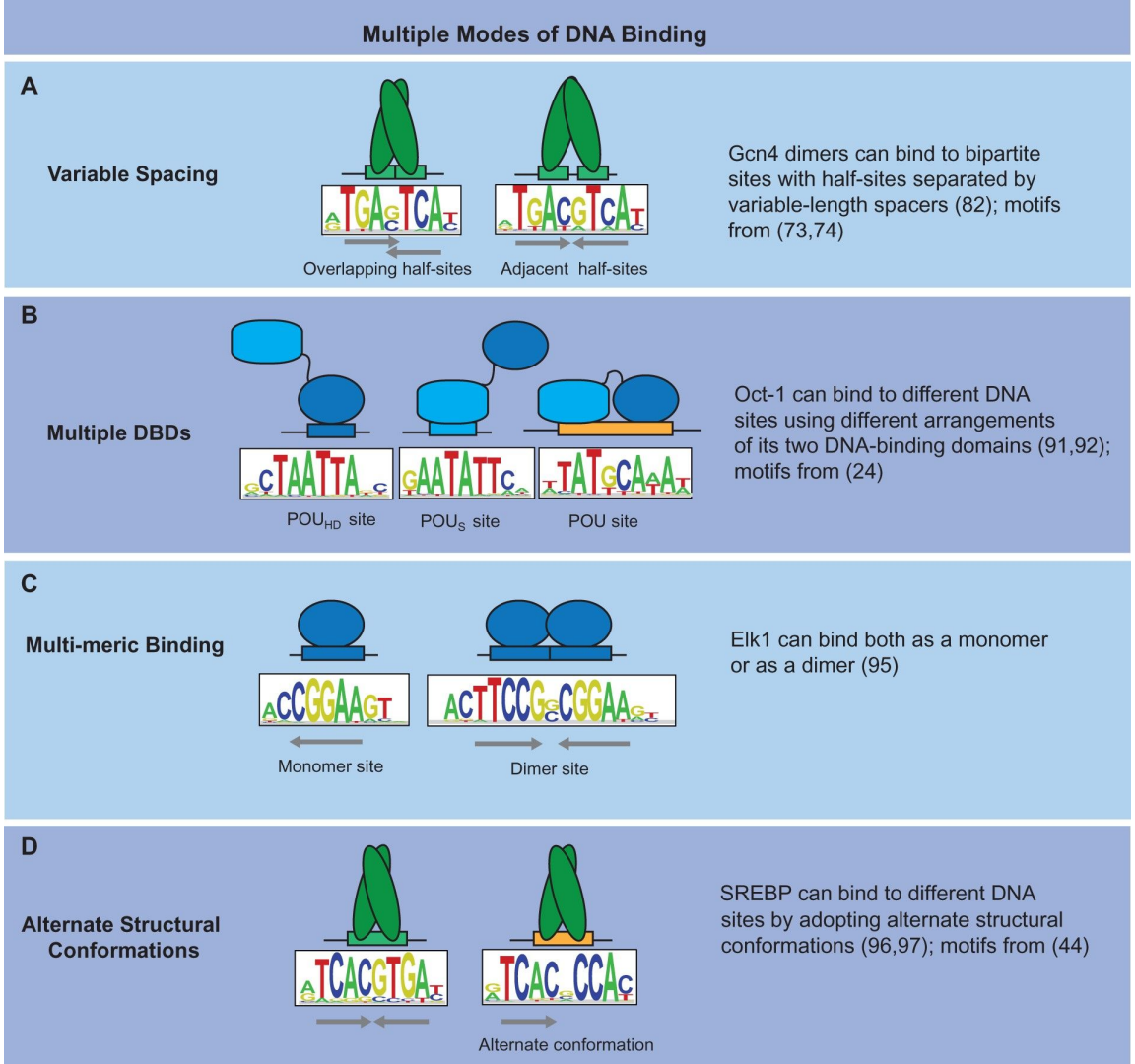
(Lambert et al., Cell 2018)

An example of TF motif degeneracy: Nuclear hormone receptors



(Lambert et al., Cell 2018)

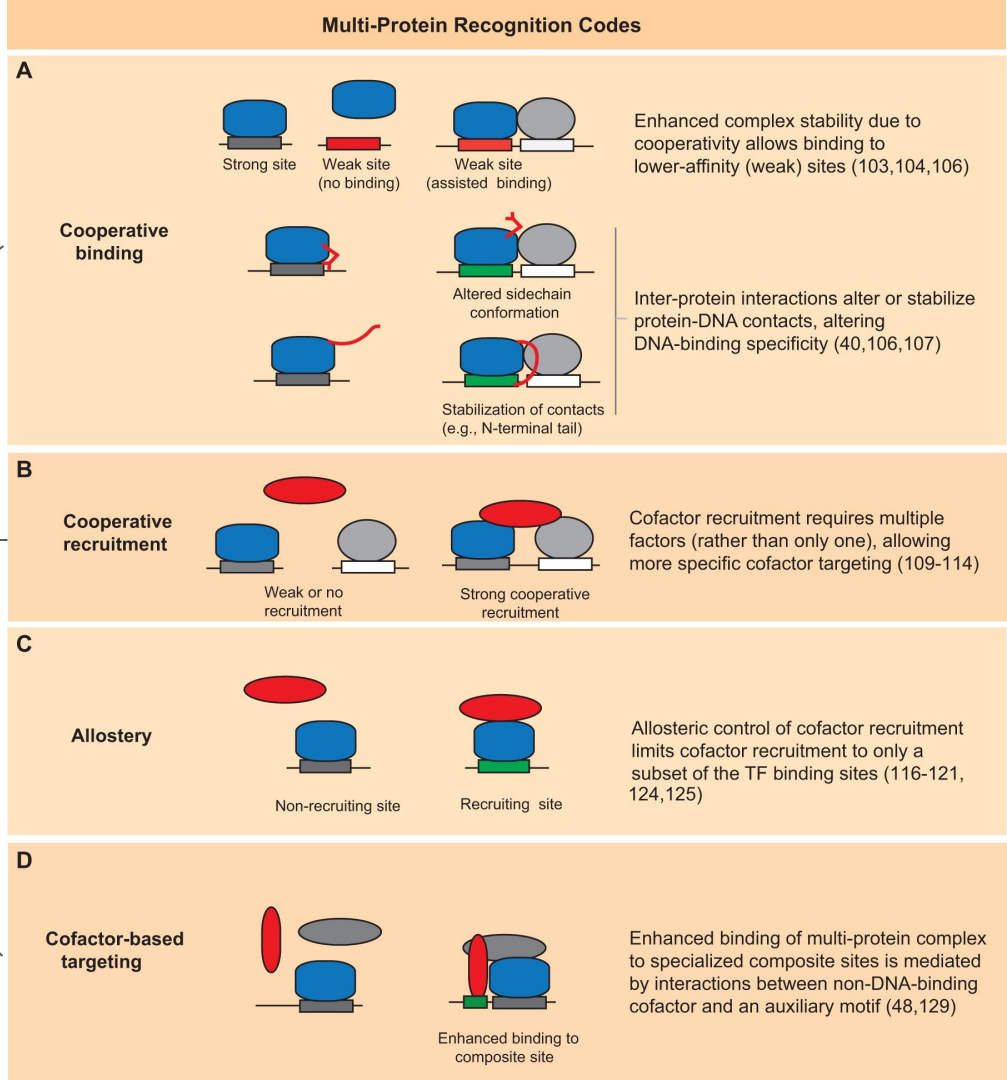
Variations in DNA binding specificity



(Siggers and Gordân, NAR 2014)

Cooperative binding

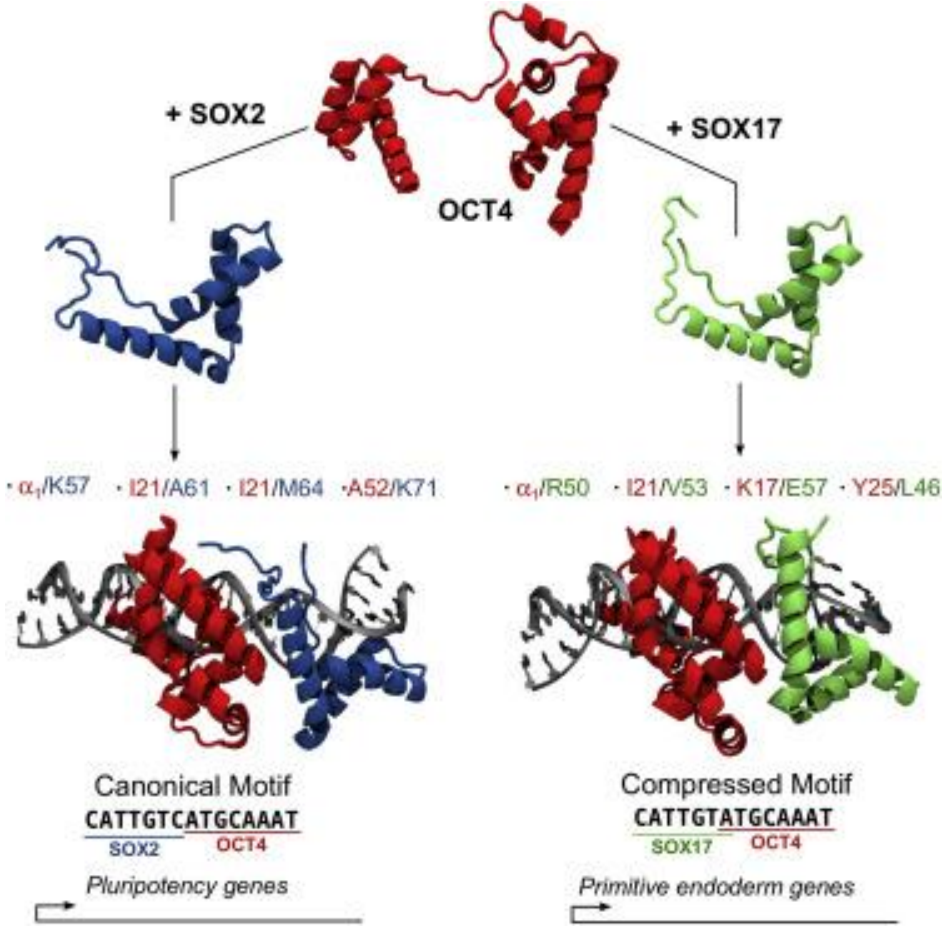
Highly combinatorial binding of TFs



(Siggers and Gordân, NAR 2014)

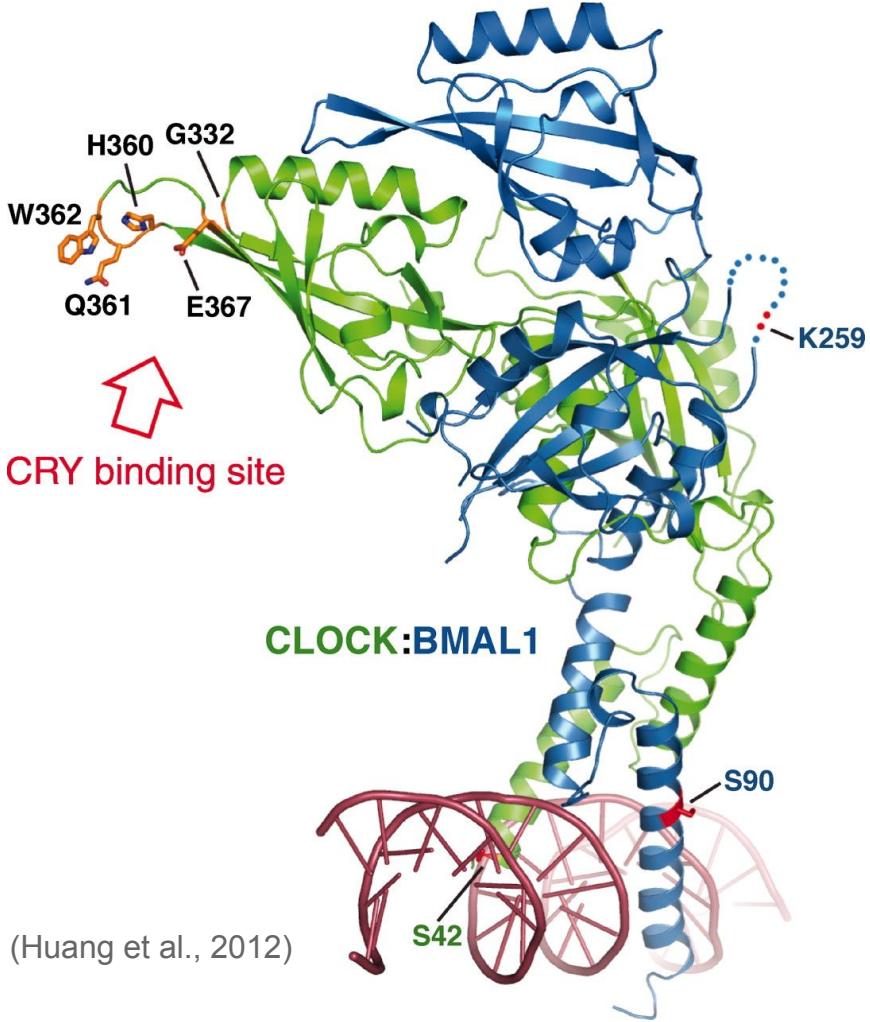
Two examples of Cooperative binding

OCT4 (POU5f1) binding upon differentiation

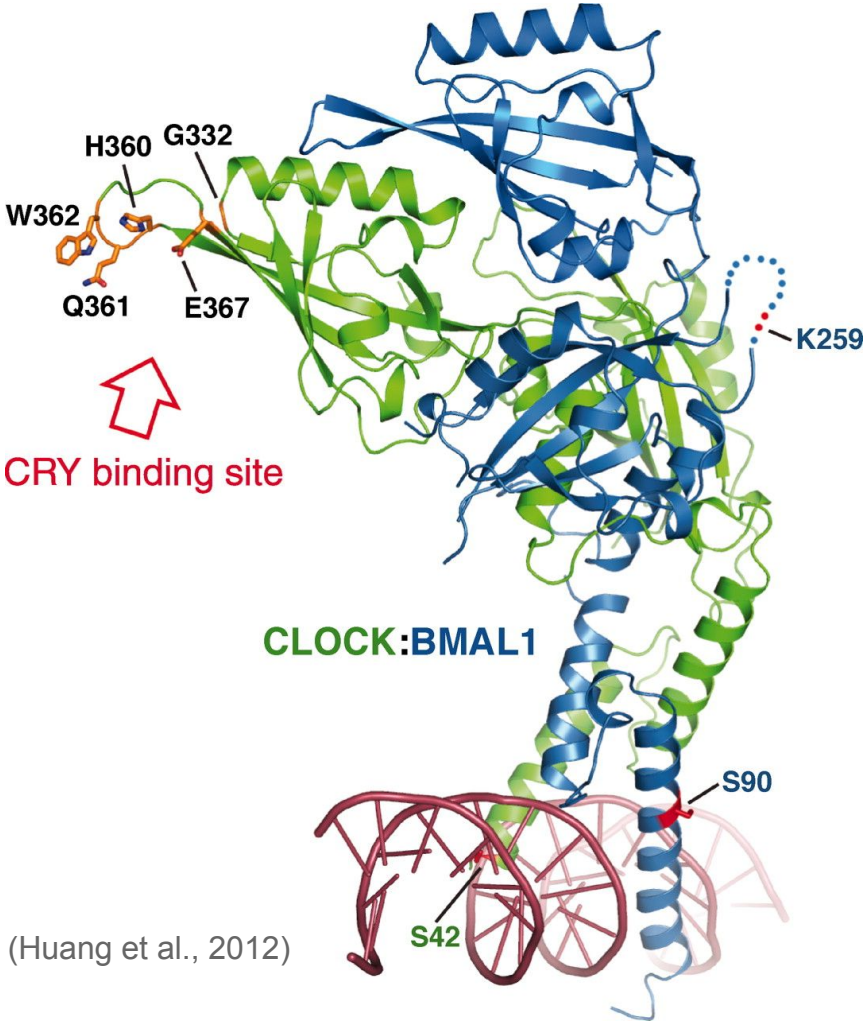


(Merino et al., *Structure* 2014)

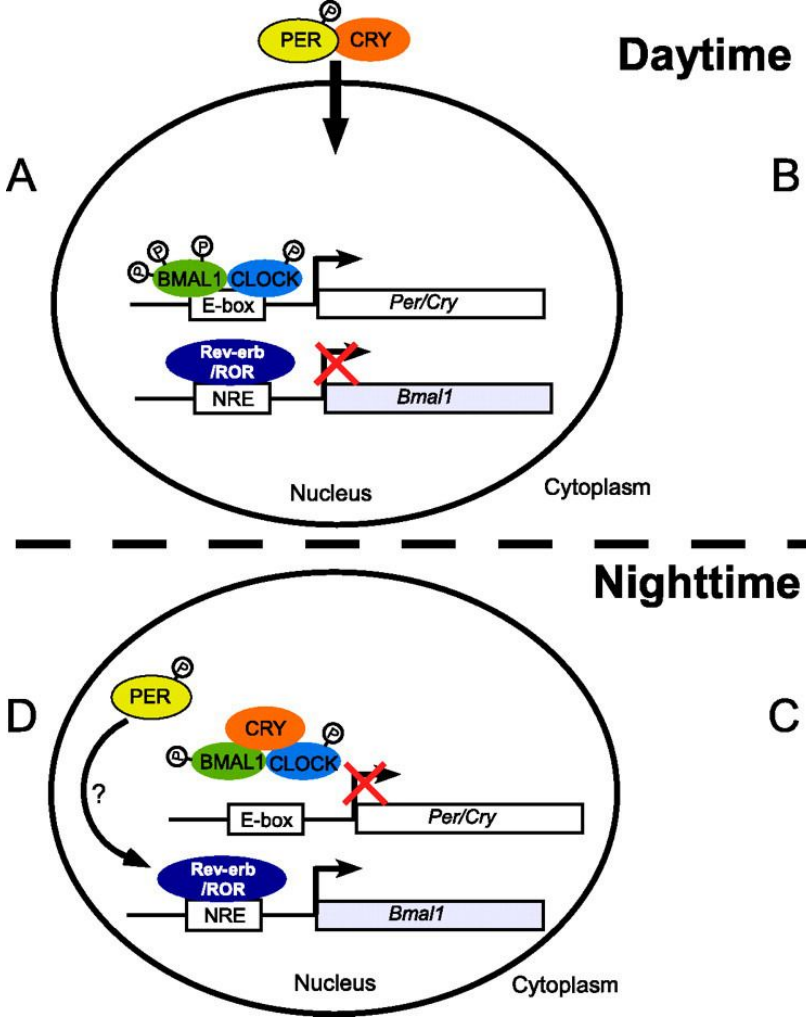
Clock-Bmal-Cry during circadian rythm



(Huang et al., 2012)



(Huang et al., 2012)

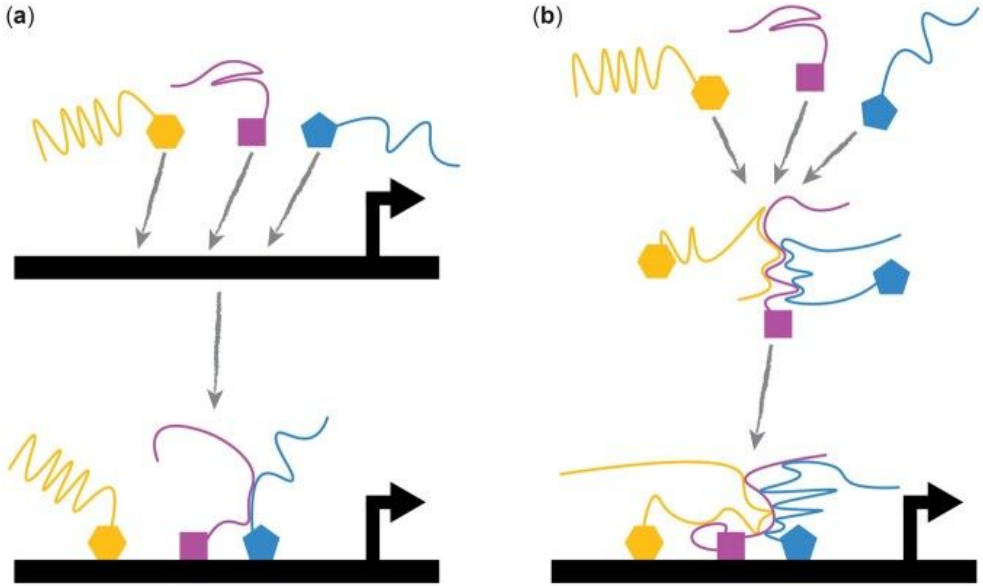


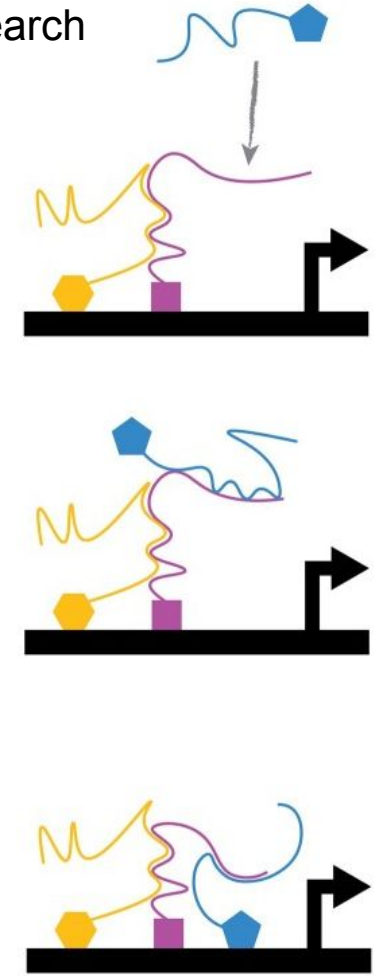
(Adapted from Ye et al., JBC 2011)

Traditional model: TFs individually go and bind to targets, then interact

Wolf pack model: TFs bind together and go together

Two-step search model:





(Adapted from Staller MV, *Genetics* 2022)

Motif analysis

- **Motif discovery** aims at finding **new** motifs that are enriched in a set of sequences (e.g. peaks) versus a background
  - Example method: MEME / STREME (Meme suite)
  - Bioconductor method: `rGADEM` package (see also the `memes` R package)

- **Motif enrichment** analysis aims at finding **known** motifs that are enriched in a set of sequences (e.g. peaks) versus a background
  - Example method: AME (Meme suite)
  - Bioconductor method: `PWMEnrich` package

- **Motif scanning** aims at finding the **occurrences of known** motifs in a set of sequences (methodologically fairly simple – which method doesn't matter much)
  - Bioconductor method: `motifmatchr`
  - (other options are the `TFBSTools` R package and FIMO of the Meme suite)

Genetic variation at TF binding sites

- Genetic variation at TF binding sites can affect the binding of the protein, and hence impact development and health
- Nevertheless, while most coding sequences show evidence of **evolutionary constraint** (e.g. purifying selection), only a small fraction of TF binding sites (11.6% of footprints) show evidence of constraint – the vast majority appears to be evolving neutrally

(Vierstra et al., Nature 2020)

- This suggests a degree of (at least partial) redundancy between regulatory elements

Assignment

- Choose a transcription factor, e.g. CREB1, REST, GATA5, EGR1, GCR
  (or any of your choice that has a motif and available ChIPseq data)
- Download the peaks for that factor (whatever organism/cell type, just make sure you use the corresponding genome!)
- Identify the instances of the factor's motif
- Answer the following questions:
  - Of all the peaks, what proportion contains a motif for the factor?
    - Expected form of an answer: of the XX peaks, XX (XX%) contain a motif
  - Of all instances of that motif in the genome (or in one chromosome), what proportion is bound by the factor (i.e. has a peak)?
    - Expected form of an answer: of the XX motif instances, XX (XX%) overlap a peak

Don't forget to *render* your markdown and push it as `assignment.html` !